

PlacedOn — Fair Assessment Model: Understanding a Human Without Encoding Bias

July 16, 2026

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PlacedOn — Fair Assessment Model: Understanding a Human Without Encoding Bias

The goal you set: a model that understands a person across many criteria and produces a report — built for India first. The trap we just stepped around: scoring people on college tier / environment / “psychology.” This plan builds the version that is powerful **and** that catches the talent pedigree-hiring misses — which is the whole business.

0. The one principle everything hangs on

| Context is a confounder you control FOR (to be fair). It is never a feature you score people ON.

You capture environment/opportunity only to (a) recognize *achievement relative to opportunity* and (b) *audit your own pipeline* for bias. You never use it to down-rank the under-resourced or up-rank the pedigreed. Assessment judges **demonstrated evidence** — what the person showed they can do — full stop.

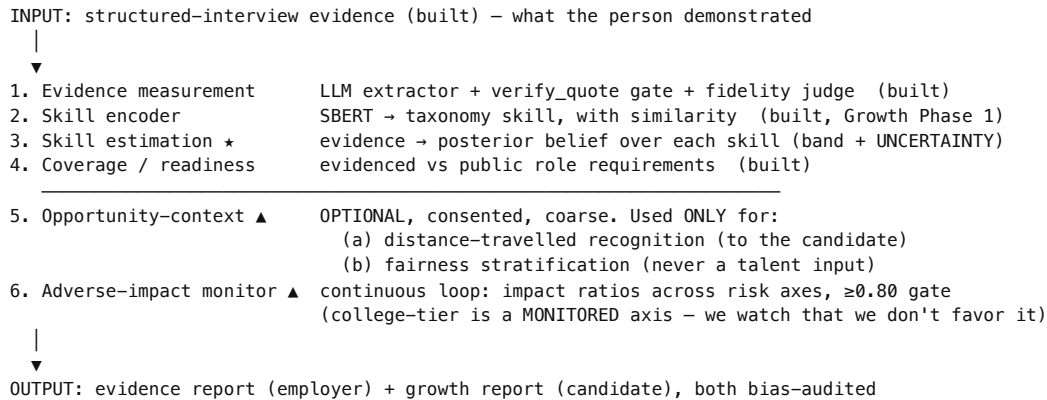
Why this is also the moat: if you score up IIT/pedigree, you’ve rebuilt the résumé filter every recruiter already has, and PlacedOn has no reason to exist. The differentiated, fundable product is the one that surfaces the capable Tier-2/Tier-3 candidate the pedigree filter drops. Fair *is* the wedge.

1. What the research actually says (grounding, not vibes)

- **The bias is real and measurable in India.** A Chennai résumé-audit field experiment found low-caste applicants must send ~20% more résumés than high-caste applicants for the same callback — and it’s taste-based, not “statistical.” *Implication: our pipeline must be continuously tested that it does not reproduce this gap.*
 - **The fair way to use context = “distance travelled.”** UK contextual admissions judge *achievement relative to opportunity* — the same grade means more from someone who reached it with less scaffolding. Context widens access; it is not a bonus for the advantaged. *This is the only sanctioned use of environment.*
 - **The cautionary tale: never boil context into one number.** The College Board’s “adversity score” (Environmental Context Dashboard) was withdrawn after backlash; even its CEO admitted reducing a person’s circumstances to a single score was “problematic.” The replacement (Landscape) shows *contextual data*, never a composite person-score. *We take the lesson: no environmental score attached to a person, ever.*
 - **The #1 fair predictor is a structured interview** (Sackett et al. 2022, structured interviews $\approx$ the strongest single validity, and lower adverse impact than unstructured judgment or pedigree). *This is exactly what we already built — evidence from structured questioning. We lean into it.*
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2. Architecture — extend the pipeline, don’t bolt on a “talent net”

There is **no** monolithic neural net that reads a person and outputs a talent number. There is a chain of small, auditable estimators plus a fairness loop wrapped around all of it.



★ new math. ▲ new fairness subsystem. Everything else already exists.

3. The math / statistics / probability — done correctly

a) Skill as estimation-with-uncertainty, not a verdict. Each evidence item updates a belief about a skill. Model it Bayesian: prior over skill level → each verified quote is an observation → posterior gives a band **and a confidence interval**. We report the uncertainty, never a false-precision point score. Thin evidence → wide interval → “needs more evidence,” not a low score.

b) Coverage / readiness = severity-weighted fraction of a role’s *public* requirements evidenced (already built). Per-role, never a person rank.

c) Fairness metrics (the real statistics):

- **Impact ratio / four-fifths rule:** $\text{selection-rate}(\text{group}) / \text{selection-rate}(\text{reference}) \geq 0.80$.
- **Selection-rate parity** across each risk axis; **calibration-within-groups** (a given band means the same skill regardless of group).
- **Significance:** two-proportion z-test / Fisher’s exact for small samples before we act on a gap.
- **Matched-pair probing:** identical evidence, varied only on a proxy (name, college tier) → outputs must be identical. This is the India audit methodology turned inward on ourselves.

d) “Distance travelled” — modeled as confounder control, not a score. Opportunity context is a *stratification variable*: when auditing, we compare outcomes *within* opportunity strata so we can see if equal evidence yields equal treatment. For the candidate, it produces a *recognition annotation* (“you built this self-taught”) — never a number added to a person and never shown to an employer as a ranking. **We do not run any regression predicting talent from environment.** That regression is the discrimination engine; it does not get built.

4. The opportunity-context module (fairness-only, India-aware)

- **Captured:** optional, consented, coarse, candidate-owned — e.g. first-generation graduate, self-taught vs. formal training, access-to-resources band. **Never** caste, religion, region, or mother tongue as direct inputs.
- **Used for exactly two things:** distance-travelled recognition (candidate-facing), and fairness stratification (internal audit).
- **Hard rules:** never a talent/assessment input · never in the employer’s view as a ranking signal · never a single composite “context score” (the College Board lesson) · deletable by the candidate.

5. The fairness loop + sub-agent system (runtime and build)

Runtime loop (the “loop system” you asked for): a continuous **adverse-impact monitor** that, on every batch of reports/matches, computes impact ratios across the monitored risk axes — **including college tier, which we watch as a red flag, not reward** — flags drift, and enforces a **fairness gate (≥ 0.80)** that any ranking surface must pass before it ships to an employer. A human reviews any flagged drift; the system never silently “corrects” people.

Sub-agents (how we build and keep it honest):

- `firewall-agent` — the banned-signal guard (strips pedigree + protected proxies from ever entering assessment; extends the copilot pedigree-strip already in the code).
- `monitor-agent` — computes the impact-ratio dashboard + significance tests.
- `bias-probe-agent` — red-teams the pipeline with matched-pair inputs (Indian-name / college-tier swaps) and fails the build if outputs diverge.
- `calibration-agent` — checks band-means are equal across groups. Same build loop we just used for the Growth Report: plan → spawn component agents → verify green → **pass the fairness gate** → integrate.

6. India-first design rules

- **Banned as talent signals (hard firewall):** caste, religion, region, mother tongue, college tier/name, gender, age, photo.
- **The wedge:** prove we surface Tier-2/Tier-3 talent that pedigree hiring misses — and prove, via matched-pair audits, that we do **not** reproduce the ~20% caste callback gap.
- **Language fairness:** assess content in the candidate’s language; English fluency $\neq$ competence (this is why voice is transcribed and assessed as text only — accent/fluency are never scored).
- **Transparency:** every report line traces to the candidate’s own words; no hidden features.

7. The first step (buildable now, no hosting needed)

Ship the **fairness foundation first** — because it’s what makes the whole model credible, and it’s the thing a pedigree model can never show:

1. `fairness/firewall.py` — deterministic banned-signal guard: pedigree + protected proxies can never enter the assessment path. Fail-closed. Unit-tested with Indian-context inputs.
2. `fairness/impact.py` — the adverse-impact monitor: impact ratio, selection-rate parity, four-fifths gate, two-proportion test — with **college tier as a monitored axis**.
3. `fairness/matched_pairs.py` — a bias-probe harness: identical evidence, swapped only on name/college tier → asserts identical output. This is your proof of fairness.

Deliverable: a **fairness report** you can show any candidate, employer, or investor — “here is evidence our pipeline does not reward pedigree.” That artifact is a moat a résumé-scorer cannot copy.

8. What we do NOT build (guardrails, restated)

No talent-from-environment regression · no environmental/adversity composite score on a person · no college-tier reward · no protected-class or caste inference · no Reddit-scraped “psychology” features (unrepresentative, unvalidated — it would bake internet stereotypes into hiring) · no single blended person-score anywhere · the AI informs, a human decides.

9. Sources

- Caste-based hiring discrimination, Chennai audit (~20% callback gap): Siddique / Thorat & Attewell — <https://docs.iza.org/dp3737.pdf> SMARTPANTS PRESERVED 241 · <https://www.sciencedirect.com/science/article/abs/pii/S0927537111000807> SMARTPANTS PRESERVED 243 SMARTPANTS PRESERVED 244
- Contextual admissions & “distance travelled”: Office for Students — <https://www.officeforstudents.org.uk/publications/contextual-admissions-promoting-fairness-and-rethinking-merit/> SMARTPANTS PRESERVED 247 · HEPI — <https://www.hepi.ac.uk/2020/05/07/above-and-beyond-predictions-no-exams-presents-an-opportunity-for-innovation-in-contextual-admissions/> SMARTPANTS PRESERVED 249 · Monash “achievement relative to opportunity” — <https://www.monash.edu/about/who/equity-diversity-inclusion/staff/equitable-decision-making/achievement-relative-to-opportunity> SMARTPANTS PRESERVED 251 SMARTPANTS PRESERVED 252
- College Board adversity-score withdrawal (never a single context score): NBC News — <https://www.nbcnews.com/news/us-news/college-board-replaces-plan-sat-student-adversity-score-n1046976> SMARTPANTS PRESERVED 255 · Higher Ed Dive — <https://www.highereddive.com/news/college-board-drops-adversity-score-and-shares-its-methods/561871/> SMARTPANTS PRESERVED 257 SMARTPANTS PRESERVED 258
- Structured-interview validity & lower adverse impact: Sackett et al. (2022), *Journal of Applied Psychology* — [revised validity estimates for selection methods.](#)